

Chapter 7: The Economics Pillar in Practice — VBC Financial Modeling, APM Readiness, and the ROI of Transformation

How Healthcare Organizations Model the Financial Impact of Value-Based Care: APM Contract Analysis, Shared Savings Mechanics, Risk Stratification ROI, and the Financial Case for Transformation Investment

From Payment Architecture to Financial Practice

Chapter 6 established the payment architecture — reference-based pricing, global budgets, the AHEAD Model, Maryland's proof of concept, and Vermont's economic transformation mandate. This chapter turns from architecture to practice: how do healthcare organizations actually model the financial impact of alternative payment models, assess their readiness to enter and succeed in VBC contracts, and build the financial case for the operational investments that transformation requires?

Financial modeling for value-based care is not the same as financial modeling for fee-for-service operations. Under FFS, the financial questions are primarily about revenue optimization: how do we maximize reimbursement for the volume we deliver? Under VBC, the financial questions are fundamentally different: what is our total cost of care for our attributed population? Where is our cost higher than the benchmark, and why? What investments in care management, primary care, or SDOH programs will reduce that cost below the benchmark, producing shared savings? These are population health economics questions, not revenue cycle questions, and they require different analytical frameworks, different data infrastructure, and different management skills.

Section 1: The APM Financial Modeling Framework

Understanding Benchmarks — The Foundation of VBC Economics

Every alternative payment model is built around a benchmark: the expected cost of care for the attributed population against which actual performance is measured. Shared savings (if actual costs are below benchmark) or shared losses (if actual costs are above benchmark) are calculated relative to this benchmark. Getting the benchmark right — or negotiating favorable benchmark methodology — is often the most important financial decision in VBC contract entry.

Benchmarks can be constructed in several ways, each with different financial implications for the entering organization:

Benchmark Construction Methods — What Healthcare Leaders Must Understand

Historical spending baseline: The most common approach — the benchmark is set at the organization's own historical cost of care, typically with a trend adjustment. Advantage: organizations with above-average historical spending start with a higher benchmark, making savings easier to achieve. Disadvantage: organizations with below-average historical spending (already efficient) face a lower benchmark with less room for savings — the 'efficient provider disadvantage.'

- Vermont AHEAD context: CMMI sets global budget baselines for hospitals based on their historical Medicare FFS spending, with adjustments. Vermont's hospitals with historically lower spending may face benchmarks that limit their financial upside under AHEAD.

Regional benchmark: The benchmark is set at regional average cost of care, not the organization's own history. Advantage: eliminates the efficient provider disadvantage. Disadvantage: organizations in high-cost regions compete against a lower benchmark than their own history would produce.

- Vermont RBP context: GMCB's reference-based pricing uses Medicare rates as the external benchmark — not Vermont hospital historical prices. This is a regional/national benchmark approach that directly addresses the problem of benchmarks based on historical high prices.

Prospective benchmark: The benchmark is set at the beginning of the performance year based on projected population needs. Used in global budgets and capitation models.

Advantage: provides revenue certainty for planning. Disadvantage: risk adjustment quality determines fairness — poor risk adjustment produces winners and losers based on case mix differences, not care management quality.

- Vermont global budget context: Vermont's Act 68 global budgets for non-CAH hospitals must specify how the prospective budget is set and adjusted for patient complexity — the risk adjustment methodology is a key design question currently under development.

The Shared Savings Calculation — A Working Model

For organizations entering ACO REACH, MSSP, or similar shared savings models, the following framework captures the core financial logic. The HTR APM Shared Savings Calculator implements this model with organization-specific data inputs.

Step	Formula / concept	Practical meaning	Vermont application
1. Benchmark setting	Benchmark PMPM × attributed member months = Total benchmark spending	This is the 'budget' the organization is measured against. Every dollar of actual spending below this number is potential shared savings.	AHEAD hospital global budget = prospective annual revenue set at beginning of year. Vermont hospital knows its 'benchmark' before the year begins.
2. Performance measurement	Actual total cost of care for attributed population	Includes all Medicare Part A and Part B spending for attributed beneficiaries, regardless of where care is delivered. Out-of-network spending counts against the ACO.	AHEAD tracks total cost of care for Medicare FFS beneficiaries attributed to Vermont hospitals across all payer settings.
3. Savings calculation	Benchmark - Actual TCOC = Gross savings (or loss)	If actual spending is below benchmark, gross savings exist. If above, gross loss.	Under global budget, the hospital receives its fixed prospective payment regardless — so the 'savings' is the margin between efficient population health management and the fixed budget.
4. Minimum savings rate (MSR)	In most shared savings models, savings must exceed a minimum threshold (typically 2-3.5%) before the organization shares in them	Protects against natural year-to-year cost variation triggering shared savings or losses that don't reflect genuine performance changes.	AHEAD uses a different structure — state-level TCOC targets rather than individual ACO MSRs — but the minimum threshold concept applies at the state level.
5. Shared savings calculation	Gross savings × sharing rate = Organization's shared savings payment (or loss obligation)	Sharing rates vary by model and risk track — typically 50-75% for one-sided models; may reach 80% for full-risk models.	Vermont RBP: savings to payers from price reduction; 100% of premium passthrough per GMCB monitoring requirement.

Figure 7.1 — APM shared savings calculation framework. Source: HTR APM Shared Savings Calculator methodology; CMS ACO model documentation.

Risk Stratification — The Economic Engine of Population Health Management

Under fee-for-service, the financial value of identifying a high-risk patient is limited — you can bill more for the additional services you provide. Under value-based care, the financial value of identifying a high-risk patient before a health crisis occurs is enormous: every hospitalization prevented by a care management intervention saves the organization the full cost of that hospitalization. The math is compelling.

A care management program that prevents 50 hospitalizations in a population of 10,000 Medicare beneficiaries, at an average cost of \$15,000 per hospitalization, generates \$750,000 in avoided cost. If the organization bears downside risk (and receives upside savings) on those prevented hospitalizations, the financial return on a care management program that costs \$200,000 to operate is approximately 3.75:1 — a 275% ROI. This is the core financial logic that makes population health management economically rational under VBC and economically irrational under FFS.

Average Medicare inpatient hospital cost prevented per event \$15,000 Approximate average — varies significantly by DRG and acuity level	Average potentially preventable ED visit cost \$1,800 Including facility and professional fees; varies by acuity
ROI: Care management program preventing hospitalizations 3-5x Typical ROI range for well-designed ACO care management programs	Vermont potentially avoidable ED visits as % of total 32.3% Each prevented = \$1,800 saved under global budget / VBC arrangement

Figure 7.2 — The financial case for population health management under VBC. Sources: CMS cost data; AHS November 2025 Transformation Report.

Section 2: APM Readiness Assessment — The 30-Dimension Framework

The HTR Value-Based Care Transformation Readiness Assessment evaluates organizational readiness for APM entry across 30 dimensions in six domains. The framework is designed to identify the gaps that are most likely to prevent an organization from achieving shared savings — or most likely to produce shared losses — before a contract is signed.

The Six Readiness Domains

- Domain 1: Strategic Clarity assesses whether the organization has a clear rationale for APM participation, an appropriate governance structure, and leadership alignment on the risk appetite and timeline for VBC transition. Organizations that enter APMs without genuine executive commitment — treating VBC as a compliance exercise rather than a strategic priority — consistently underperform.
- Domain 2: Data and Technology evaluates the analytics infrastructure needed to manage a VBC population: attribution list generation, total cost of care measurement, care gap identification, risk stratification, and quality reporting automation. This domain is Vermont's most significant organizational gap — most Vermont community hospitals lack the internal analytics capability to use VHCURES data effectively for population health management. The joint AHS-GMCB analytics vendor procurement addresses the statewide gap; individual hospital analytics capacity must be built alongside it.
- Domain 3: Care Delivery Capability assesses the clinical programs needed to act on risk stratification insights: care management programs organized around risk tiers, primary care PCMH readiness, behavioral health integration, post-acute care network

relationships, community health worker deployment, and patient outreach and engagement capability. Vermont's Blueprint for Health PCMHs and CHTs provide a strong care delivery foundation; the gap is in building on this foundation to serve the more complex patients that AHEAD's population health requirements target.

Domain 4: Network and Partnerships evaluates the relationships the organization has with the specialists, post-acute facilities, and community organizations that determine whether high-risk patients receive coordinated care or fragmented care after a hospitalization or specialist referral. Vermont's Oliver Wyman-designed COE regionalization framework and CIN development address this domain systematically.

Domain 5: Revenue Cycle examines the billing and coding practices that determine whether the organization's VBC contracts are financially managed correctly: HCC coding completeness (essential for risk adjustment accuracy), RAF gap identification, attribution denial management, and VBC contract payment reconciliation. This domain is covered in depth in Chapter 9.

Domain 6: Workforce Operations evaluates the staffing model, credentialing efficiency, retention rates, and training capability that determine whether the care coordination programs identified in Domain 3 can actually be implemented and sustained.

VBC Readiness Scoring — What the Numbers Mean

Each of the 30 dimensions is scored 1-4: Pre-transition (1), Developing (2), Operational (3), Optimized (4). A total score below 60/120 indicates the organization is not ready for full-risk VBC; the gap analysis identifies the highest-priority investments before contract entry.

Vermont-specific benchmarks: Blueprint-participating PCMH practices with active CHT support typically score 8-12 in Domain 3 (care delivery) — strong by national standards. Most Vermont hospitals score 3-7 in Domain 2 (data and technology) — a critical gap given that data infrastructure is the prerequisite for population health management. The RHRC engagement's explicit focus on developing analytic dashboards for hospitals reflects the recognition that Domain 2 is the binding constraint for most Vermont providers.

Section 3: Contract Analysis — Reading an APM Contract for Financial Risk

APM contracts are complex, and the financial risk embedded in them is often not apparent from the headline terms. A shared savings contract that appears attractive at the summary level may contain provisions — benchmark methodology choices, quality withhold structures, attribution methodology, carve-out provisions — that substantially change the financial calculus. Every organization entering a significant APM contract should apply the following analysis framework.

The 65-Item VBC Contract Review Checklist: Key Categories

The HTR VBC Contract Review Checklist (available in the Implementation Toolkit) covers 65 specific contract provisions in eight categories. The most financially consequential categories are:

Contract provision category	Key questions and financial risk
Benchmark methodology	How is the benchmark set? Historical? Regional? Blended? What trend adjustment is applied? Is there a minimum savings rate, and what is it? What happens if the benchmark is reset mid-contract — does the organization benefit from demonstrated performance improvements or start over?
Attribution methodology	How are patients attributed — prospectively (you know your panel before the year) or retrospectively (you find out after

	the year)? What is the attribution methodology — plurality of primary care visits? Most costly provider? Attribution methodology determines your denominator and your exposure.
Quality withhold	What percentage of shared savings is withheld pending quality performance? What measures are used? Are they measures the organization currently tracks with reliable data? A 30% quality withhold on 50% of gross savings means quality performance determines 15% of total financial outcome — understand the measures before signing.
Risk corridors and stop-loss	Under full-risk models, is there a stop-loss provision that caps the organization's downside exposure? What is the per-member-per-month or aggregate stop-loss threshold? For organizations entering full-risk for the first time, stop-loss provisions are essential financial protection.
Carve-outs	What services are excluded from the shared savings calculation? High-cost drugs? Mental health carve-outs? Specialty carve-outs? Each carve-out reduces the organization's ability to manage total cost of care and reduces the savings opportunity.
Reconciliation timing	When are shared savings or losses settled — annually, semi-annually, quarterly? Reconciliation timing affects cash flow planning and the organization's ability to invest shared savings in care improvement programs.

Section 4: Hospital Financial Modeling Under Global Budgets — The Vermont Framework

For Vermont hospitals, the most operationally relevant financial modeling challenge is not APM shared savings calculation — it is global budget management. As Act 68's mandatory global budgets take effect for non-CAH hospitals beginning FY2028, hospital CFOs need to transition from FFS revenue optimization to global budget management. These are fundamentally different financial disciplines.

The Global Budget Financial Management Model

Under a global budget, a Vermont hospital's annual revenue is set prospectively. The financial management objective shifts from maximizing volume and revenue to achieving the highest quality population health outcomes within the fixed revenue envelope. The key financial metrics change accordingly:

Under FFS — optimize these metrics	Under global budget — optimize these metrics
Revenue per discharge — maximize reimbursement per case	Cost per discharge — minimize unnecessary cost while maintaining quality
Admission volume — more admissions = more revenue	Avoidable admission rate — every preventable admission is a cost, not revenue
Length of stay — maximize revenue per stay	Efficient length of stay — unnecessary days are costs without revenue offset
Service line volume — grow profitable service lines	PMPM population health cost — manage population health to stay within budget
Payer mix — maximize commercially insured cases	Quality performance — quality metrics determine budget adjustments and AHEAD performance

Denials management — recover denied claims	Potentially avoidable utilization — prevent the ED visits and admissions that waste budget
Case mix index — capture higher-acuity cases	Community investment ROI — SDOH programs that reduce utilization have direct financial return
Operating margin per adjusted discharge	Operating margin within global budget — sustainability within fixed revenue

Figure 7.3 — Financial metrics transformation: FFS versus global budget environment. Source: HTR Economics pillar framework.

The Hospital Financial Stress Test Model

The HTR Hospital Financial Stress Test Model — available in the Research Lab for Professional and Advisory subscribers — allows hospital CFOs and finance teams to model their organization's financial position under different payment reform scenarios. The model's key scenarios for Vermont hospitals in 2026:

- RBP scenario: What does our financial position look like if commercial prices are set at 200% of Medicare (Oliver Wyman's recommended benchmark) or 250% (GMCB's phased approach)? What service lines have prices furthest above the benchmark and face the most revenue reduction?
- Global budget scenario: Under a prospective revenue cap equal to our current net patient revenue minus X%, what is our operating margin? What cost reductions or service redesigns are required to maintain solvency?
- Medicaid cut scenario: With H.R. 1's Medicaid revenue reductions phasing in from 2027-2031, what is our projected total revenue under conservative, base, and optimistic assumptions about our Medicaid payer mix?
- Transformation investment scenario: If we invest \$X million in care management, CHW deployment, and telehealth, what reduction in avoidable hospitalizations and ED visits is required to achieve positive ROI within our VBC contract or global budget framework?

Section 5: The ROI of Transformation — Building the Financial Case

Every transformation investment — a care management program, a behavioral health integration initiative, an SDOH screening program, a telehealth platform — requires a financial justification. Under fee-for-service, this justification is often difficult to construct because the investment may reduce utilization (and revenue) even when it improves outcomes. Under value-based care and global budgets, the financial case for transformation investment is considerably stronger.

The Transformation ROI Framework

The following framework structures the financial case for transformation investments in a VBC environment. It is the analytical backbone of HTR Advisory engagements that involve transformation investment decisions.

Investment type	Revenue/savings mechanism	Calculation approach	Vermont example
Care management program	Prevents hospitalizations and ED visits that, under VBC, are costs without offsetting revenue	(Prevented events) × (average cost per event) - (program operating cost) = Net savings. Divide by program cost for ROI.	Vermont: 32.3% of ED visits potentially avoidable. CHT program preventing 500 avoidable ED visits/year at \$1,800 each = \$900K savings. If CHT costs \$300K/year, ROI = 200%.

Behavioral health integration (CoCM)	Reduces downstream high-cost utilization — hospitalizations, ED visits for MH crisis — for patients receiving earlier MH intervention	Estimate reduction in MH-related ED visits and hospitalizations for CoCM-managed patients vs. usual care. Vermont: 245 suicide/SH ED visits per 10,000 — each prevented = \$1,800 saved plus downstream cost avoidance.	Vermont Blueprint MHI pilot: 80% of administrative entities increased CHT staffing. Every MH-related hospitalization prevented at \$15,000 average saves 50x the cost of a CoCM BHCM monthly salary cost.
SDOH program investment	Reduces avoidable utilization driven by unmet social needs — housing-related ED visits, medication non-adherence from food insecurity, missed appointments from transportation barriers	Literature suggests 5-15% reduction in total cost of care for high-SDOH-burden populations receiving coordinated SDOH intervention. Apply to Vermont's identified high-SDOH populations.	Blueprint HRSN screening + CHT navigation: SDOH intervention for the 9% of Vermonters who are food insecure, if it reduces avoidable hospitalizations by even 10%, produces system savings exceeding program cost at scale.
Primary care investment (PCMH enhanced payments)	Reduces hospitalizations and specialty care through better chronic disease management and early intervention	Blueprint ROI study: \$5.8M reduction in medical expenditure per \$1M invested. At this ratio, \$3M in Blueprint PCMH investment reduces system costs by \$17.4M.	Vermont AHEAD primary care investment mandate: EAST Fund investment in primary care is essentially a system investment with documented 5.8:1 ROI in Vermont's own experience.
Telehealth for rural access	Reduces transportation-related care avoidance and enables earlier intervention before crisis; reduces specialist transport costs for rural hospitals	Travel cost savings + avoided emergency transport + reduced length of stay for patients manageable via telehealth specialty consultation. RPM: each prevented CHF hospitalization saves \$15,000.	Vermont RHT Program RPM grants: CHF readmission prevention via RPM in a population of 200 high-risk patients, preventing 10% of 30-day readmissions at 30% readmission rate, saves \$90,000/year — ROI positive at \$10,000/year RPM program cost.

Figure 7.4 — Transformation investment ROI framework with Vermont examples. Sources: HTR Economics pillar framework; Blueprint for Health ROI study; CMS cost data; Vermont data from AHS Transformation Reports.

Section 6: Making Care Primary — Vermont's Primary Care Economics

The AHEAD Model's requirement that states demonstrate increased investment in primary care as a percentage of total spending is not just a quality metric — it is an economics imperative. The foundational economic argument for primary care investment in a global budget environment is that primary care is the highest-return investment a healthcare system can make: the same \$1 invested in primary care produces greater total cost of care reduction than \$1 invested in hospital care or specialty care.

Vermont's Blueprint for Health embeds this economic logic in its design. The all-payer PMPM payments to PCMH practices are structured as an investment in reduced downstream costs — and Vermont's own evaluation data confirms the 5.8:1 ROI. Under AHEAD, primary care investment will be measured as a percentage of total spending,

creating an explicit financial accountability framework for the primary care investment argument.

For Vermont hospitals operating under global budgets, the financial relationship between primary care investment and hospital cost management is direct: a better-functioning primary care system means fewer hospital admissions from preventable chronic disease exacerbations, fewer avoidable ED visits, and fewer hospitalizations for conditions that should have been managed in outpatient settings. The hospital that invests in primary care strengthens the community health infrastructure that reduces its own inpatient costs — a virtuous cycle that fee-for-service can never produce but global budgets require.

Key Concepts

Term	Definition
Benchmark	The expected cost of care against which actual APM performance is measured. Shared savings or losses are calculated relative to the benchmark. Benchmark methodology — historical vs. regional, trend adjustment, risk adjustment — is the most consequential financial design choice in APM contract entry.
Attribution	The process of assigning patients to an ACO or provider for purposes of shared savings/loss calculation. Prospective attribution (known before the year) enables proactive care management; retrospective attribution (known after the year) creates uncertainty. Attribution methodology determines the denominator and exposure.
Shared savings rate	The percentage of savings below benchmark that the organization receives. Typically 50-75% for one-sided risk models; may reach 80% for full-risk models.
Minimum savings rate (MSR)	The threshold below which savings are not shared — typically 2-3.5% of benchmark — protecting against natural year-to-year variation triggering unearned payments.
Risk corridor	A provision in full-risk APM contracts that caps the organization's maximum financial loss at a defined threshold, providing stop-loss protection for organizations bearing downside risk.
RAF (Risk Adjustment Factor)	The multiplier applied to a benchmark to account for the health status of the attributed population. Accurate HCC (Hierarchical Condition Category) coding is essential for RAF accuracy. RAF gaps — diagnoses that should be coded but are not — create benchmarks that understate the population's expected cost, producing apparent savings that disappear when coding is corrected.
Potentially avoidable utilization (PAU)	Hospitalizations, ED visits, and readmissions that could have been prevented through effective primary care, care management, or SDOH intervention. The core financial target for care management investment under VBC and global budgets.
VBC Transformation Readiness Assessment	The HTR 30-dimension, 6-domain assessment evaluating an organization's readiness to enter and succeed in value-based care contracts. Total score below 60/120 indicates pre-transition status; gap analysis identifies highest-priority investments before contract entry.

Sources: CMS ACO model documentation; HTR Economics pillar framework; Vermont Blueprint for Health ROI study (Population Health Management); GMCB hospital budget data; Oliver Wyman Act 167 Report; AHS Transformation Reports; Vermont AHEAD State Agreement; Act 68 of 2025.